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E1 Evidence: Final Report

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MR2004B.602: Implementation of Mechatronic Systems

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Introduction

This project consisted of the design and implementation of a prototype capable of moving a solenoid along a horizontal axis through a belt-driven mechanism. The system was designed to allow controlled linear motion over a distance of approximately 40 cm. A NEMA 17 stepper motor with an integrated pulley was used to drive the belt, while a passive pulley on the opposite side maintained tension and direction. The entire system was controlled through an Arduino Uno using an A4988 driver and a joystick module, enabling precise manual control of the solenoid's position.

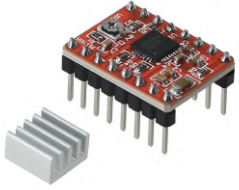
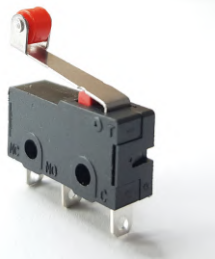
The main objective of the project was to create a functional “Goalkeeper” prototype capable of competing against other similar systems in a game board. The aim was to prevent a ping-pong ball from crossing the goal line by quickly repositioning the solenoid to block incoming shots. In addition to its movement capabilities, the solenoid could be activated via a push button, generating a mechanical force of approximately 20 N, strong enough to hit and deflect the ping-pong ball away from the goal area.

Throughout the development of this project, we applied and reinforced a variety of skills and concepts acquired during the course. In terms of mechanical design, we use SolidWorks for 3D modeling, which allowed us to create detailed and functional parts, such as the supports for the motor and pulleys, as well as a custom 3D-printed casing for the solenoid. The case included a movable part to efficiently transfer the mechanical impact produced by the solenoid.

On the electronics side, we deepened our understanding of Arduino programming, both in writing code to control all the components and in configuring the necessary electronics to interface sensors, actuators, and power components. Additionally, we applied key concepts in basic electronics, particularly the use and behavior of components such as diodes, resistors, capacitors, and transistors, which were essential for protecting the circuit and managing current flow.

Proposal design

Component	Description / Specs	Quantity	Image
NEMA 17 Stepper Motor	1.8° step angle, 12V	1	

A4988 Stepper Driver	For NEMA 17 control	1	
Arduino Uno	Microcontroller	1	
Joystick Module	2-axis analog	1	
Solenoid	12V, 20 N force	1	
Push Button	Momentary switch	1	
Limit switches	Limits the range of motion	2	
Power Supply	24V, 5A (for the motor and solenoid)	1	

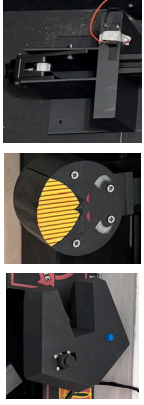
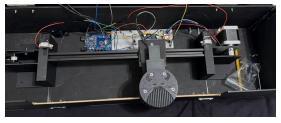
Batteries	1.5V (for the Arduino)	4	
PLA Filament	For 3D printing	-	
3D-printed Parts	Supports, casing, controller	-	
Buck converter LM2596	DC-DC step-down voltage regulator module	1	
Misc. Electronics	Wires, resistors, mosfet, capacitor	-	

Table 1. Components

Proposal Validation

To validate our project proposal from an engineering standpoint, we used a combination of design documentation, simulation, prototyping, and performance evaluation. This ensured the feasibility and robustness of our manually controlled goalkeeper and allowed us to compare its advantages against other types of solutions.



Figure 1.1 Solenoid Case CAD

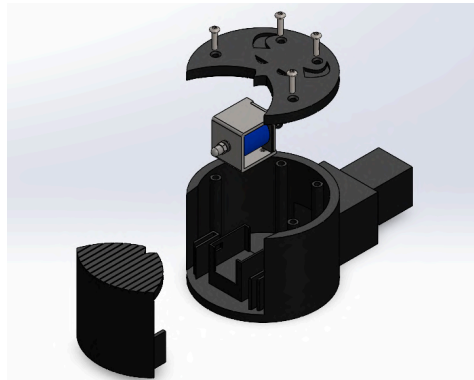


Figure 1.2 Exploded view of the Solenoid and Case CAD

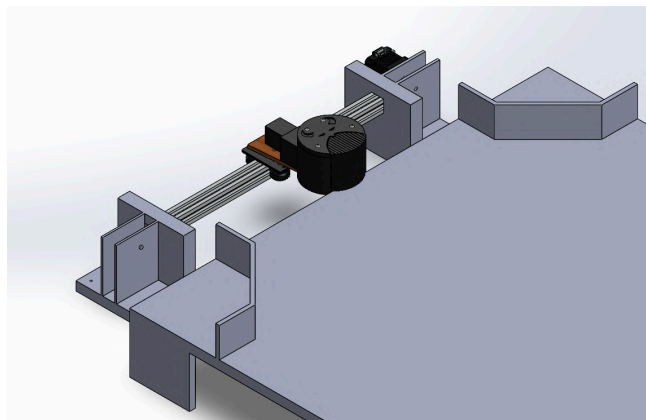


Figure 1.3 Final Design CAD

Our system uses a manual open-loop control:

- The user interacts with a joystick or push-buttons connected to an Arduino Uno.
- The Arduino interprets the input and drives a DC motor (or servo motor) to rotate the goalkeeper.
- Limit switches are used to stop motion at each extreme, preventing mechanical damage.

Even though it's an open-loop system, the inclusion of limit switches provides boundary safety, ensuring reliability despite the absence of feedback sensors.

To experimentally validate the system, we carried out several performance tests. First, we verified that the motor responded immediately and correctly to user input, ensuring smooth and controllable motion. Then, we tested the reliability of the limit switches under repeated activation, confirming that they consistently halted the motor at the correct positions. Additionally, we assessed the structural strength of the MDF components, hinges, and mounts to ensure that the goalkeeper could withstand impacts from a moving ball without loosening or failing.

Our solution presents several competitive advantages over existing alternatives that rely on sensors or autonomous motion. One key benefit is cost-effectiveness, as we eliminated the need for multiple sensors or complex control algorithms. Additionally, our approach increases interactivity, since users are directly involved in the gameplay rather than observing automated responses. The system is also easier to assemble, program, and maintain, making it ideal for educational settings where hands-on learning is prioritized. Unlike fully automated systems that can be difficult to troubleshoot or modify, our design remains accessible and adaptable.

The results from our tests confirmed our initial hypothesis: a manually controlled goalkeeper system, regulated with limit switches and powered by a motor through Arduino programming, can effectively simulate real goalkeeping behavior in a small-scale game. It provides a responsive and durable solution that balances user engagement with simplicity and technical reliability.

Proposal Implementation

Here is a [video](#), showing our goalkeeper prototype fully built and operating as expected. The structure was stable, the mechanical movement was smooth, and the goalkeeper successfully responded to user inputs through the controller. The limit switches worked correctly, stopping the motor at the designed endpoints and protecting the system from over-rotation. Overall, the project functioned just as we had planned and validated in our initial design. However, during final testing and gameplay, we realized that increasing the motor speed would have improved the user experience and gameplay dynamics. While it did not affect the core functionality, adding speed variation could make the goalkeeper more responsive.

Conclusion

This project provided a comprehensive opportunity to integrate mechanical design, electronics, and programming concepts into a cohesive and functional interactive goalkeeper prototype. The process required effective collaboration and teamwork, allowing us to leverage each member's strengths and foster a productive working environment. By combining theoretical knowledge with practical application, we successfully achieved our core objectives, developing a responsive and reliable system that performed consistently in line with our original design specifications.

Throughout the development phases, we deepened our understanding of how various engineering disciplines intersect, from mechanical movement and sensor integration to control algorithms and software responsiveness. The hands-on experience highlighted the importance of iterative testing and refinement, as well as troubleshooting challenges that arose during implementation. For example, although the system functioned as intended, we recognized opportunities to enhance the prototype further, such as incorporating adjustable speed control to improve the realism and adaptability of the gameplay experience.

Moreover, this project served as a valuable platform for enhancing essential skills beyond the technical scope. Effective communication, project planning, and adaptability were critical in navigating unforeseen obstacles and coordinating complex tasks among team members. All this combined with our technical growth, have better prepared us for future projects.

Individual Contributions

Rebeca:

In this project, my individual contribution focused on the mechanical design, CAD modeling, physical assembly, and documentation. I worked on building the structure using MDF, aligning and assembling components like the base and hinges, and ensuring the physical system functioned properly. I also helped develop part of the CAD model to visualize the design and contributed to writing the final report and organizing team efforts. I did not participate in the programming or electronic wiring, but I stayed involved in ensuring those elements integrated well with the mechanical build. Compared to last semester, I strengthened my understanding of how mechanical components come together in a real prototype, and I developed soft skills such as team coordination, adaptability, and time management. As a team, we successfully created a functional goalkeeper prototype. For future development, we could improve the control system and enhance portability.

Sofia:

During this project, I contributed significantly by researching and purchasing most of the components and materials we used. I also participated in the 3D printing and laser cutting of the MDF for the box and was actively involved in the complete mechanical assembly of the project. This included tasks such as installing the hinges on the box, screwing the base, and performing any necessary drilling. In addition to the hands-on work, I also contributed to the project documentation and provided evidence demonstrating that the system was functioning as intended. Compared to last semester, I believe I have developed a wide range of new skills. From improving my familiarity with tools such as laser cutters and drills, developing better mechanical assembly techniques, having a more structured project planning and documentation of the progress. As an individual and as a team I dear to say that this

experience has not only enhanced our technical and management skills, but also created an operational goalkeeper. For future iterations of the project we could improve its movement, use better materials for the box to enhance its durability.

Montserrat:

In the development of this project, I contributed to the initial design of the goalkeeper and the SolidWorks for the initial goalkeeper assembly. As well as the case and support design, and the model in SolidWorks. Additionally, I participated in the 3D printing and the laser cutting for the necessary pieces for the assembly. I also helped throughout the whole building process of the goalkeeper including the drilling, screwing, cutting and assembling. Finally, I participated in the documentation throughout the challenge deliverables. Overall, this project made me learn a lot throughout the semester. Among the things I learned was using SolidWorks, and I learned to use different tools from the labs, like laser cutters and 3D printers, in order to build the goalkeeper. I also learned a lot from the code made for programming the project, it helped me understand more about how Arduino programming works. Overall, I developed teamwork skills, and this experience gave me new knowledge. For a future development of the project, we could improve the design and movement, as well as use better materials.

José Pablo:

Throughout the development of this project, I was deeply involved in both the design and implementation stages. One of my main contributions was the CAD modeling of the solenoid case, which I designed to securely hold the solenoid and include a movable part to transfer the impact. I also designed the CAD model of the controller, which housed the joystick and the button, allowing for intuitive and comfortable manual control of the system. These designs were carefully created using SolidWorks and later 3D printed to become part of the final prototype.

In addition to the mechanical components, I was responsible for the electronic implementation and programming. I connected and configured all electronic components, including the Arduino Uno, A4988 driver, joystick, and solenoid. I wrote the Arduino code that controlled both the stepper motor for smooth linear motion and the activation of the solenoid through a button press.

This project allowed me to apply and strengthen my skills in 3D design, electronics, and embedded programming. It was a challenging and rewarding experience that taught me how to integrate multiple disciplines to create a fully functional and interactive electromechanical system.

References

Hyperlaps ©. (2022, September 15). *Innovative DIY (Do it Yourself) board game* -

Hyperlaps.io © [Video]. YouTube. <https://www.youtube.com/watch?v=JqSubWfrhvw>

Annex

Goalkeeper Code

```
const int dirPin = 2;
const int stepPin = 3;

int xPin = A0;
int xVal;
int ButtonPin = 10;
int ButtonState = 0;
int MosfetPin = 9;
int LeftSwitchPin = 7;
int RightSwitchPin = 6;
int LeftSwitchState = 0;
int RightSwitchState = 0;
int dt = 50;

int stepsPerRevolution = 200;
int stepDelay = 2;

void setup() {
  pinMode(dirPin, OUTPUT);
  pinMode(stepPin, OUTPUT);
  pinMode(xPin, INPUT);

  pinMode(ButtonPin, INPUT_PULLUP);
```

```

pinMode(MosfetPin, OUTPUT);

pinMode(LeftSwitchPin, INPUT_PULLUP);
pinMode(RightSwitchPin, INPUT_PULLUP);

digitalWrite(MosfetPin, LOW);

Serial.begin(9600);
}

void loop() {

    ButtonState = digitalRead(ButtonPin);

    // Control del solenoide
    if (ButtonState == LOW) {
        digitalWrite(MosfetPin, HIGH);
    } else {
        digitalWrite(MosfetPin, LOW);
    }

    xVal = analogRead(xPin);
    LeftSwitchState = digitalRead(LeftSwitchPin);
    RightSwitchState = digitalRead(RightSwitchPin);

    // Depuración
    Serial.print("X value = ");
    Serial.print(xVal);
    Serial.print(" - Left Switch = ");
    Serial.print(LeftSwitchState);
    Serial.print(" - Right Switch = ");
    Serial.print(RightSwitchState);
    Serial.print(" - Button = ");
    Serial.println(ButtonState);

    // Control del motor a pasos basado en el joystick
    if (RightSwitchState == HIGH && xVal < 450) {
        digitalWrite(dirPin, HIGH); // Derecha
        for (int i = 0; i < 100; i++) {
            digitalWrite(stepPin, HIGH);
            delayMicroseconds(650);
            digitalWrite(stepPin, LOW);
            delayMicroseconds(650);
        }
    }
    else if (LeftSwitchState == HIGH && xVal > 650) {
        digitalWrite(dirPin, LOW); // Izquierda
        for (int i = 0; i < 100; i++) {

```

```

digitalWrite(stepPin, HIGH);
delayMicroseconds(650);
digitalWrite(stepPin, LOW);
delayMicroseconds(650);
}
}
}

```

Attachments

[Goalkeeper_Evidence - Google Drive](#)

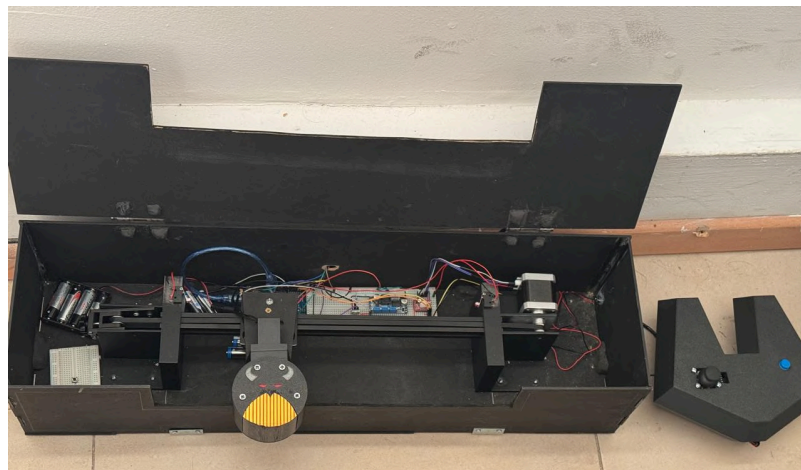


Figure 2.1 Final Prototype



Figure 2.2 Final Prototype

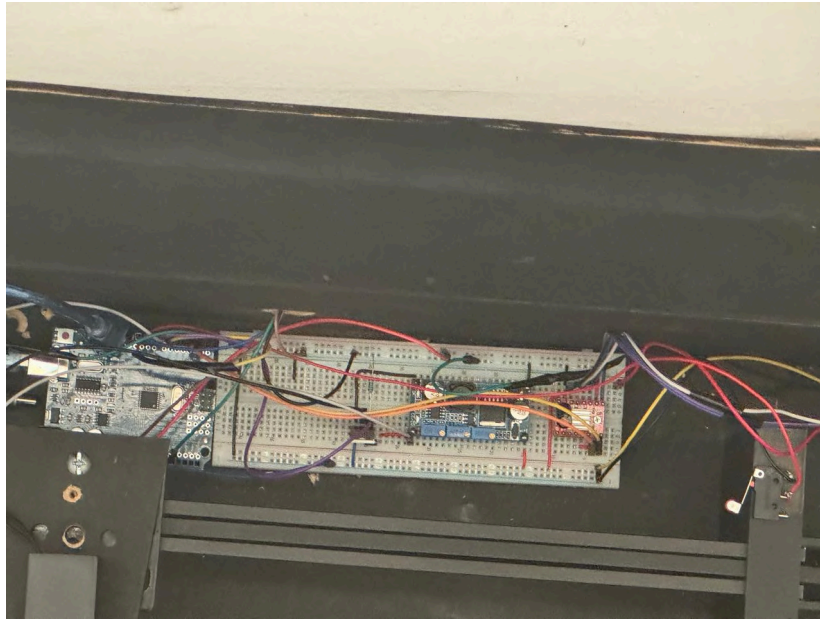


Figure 2.3 Electronics

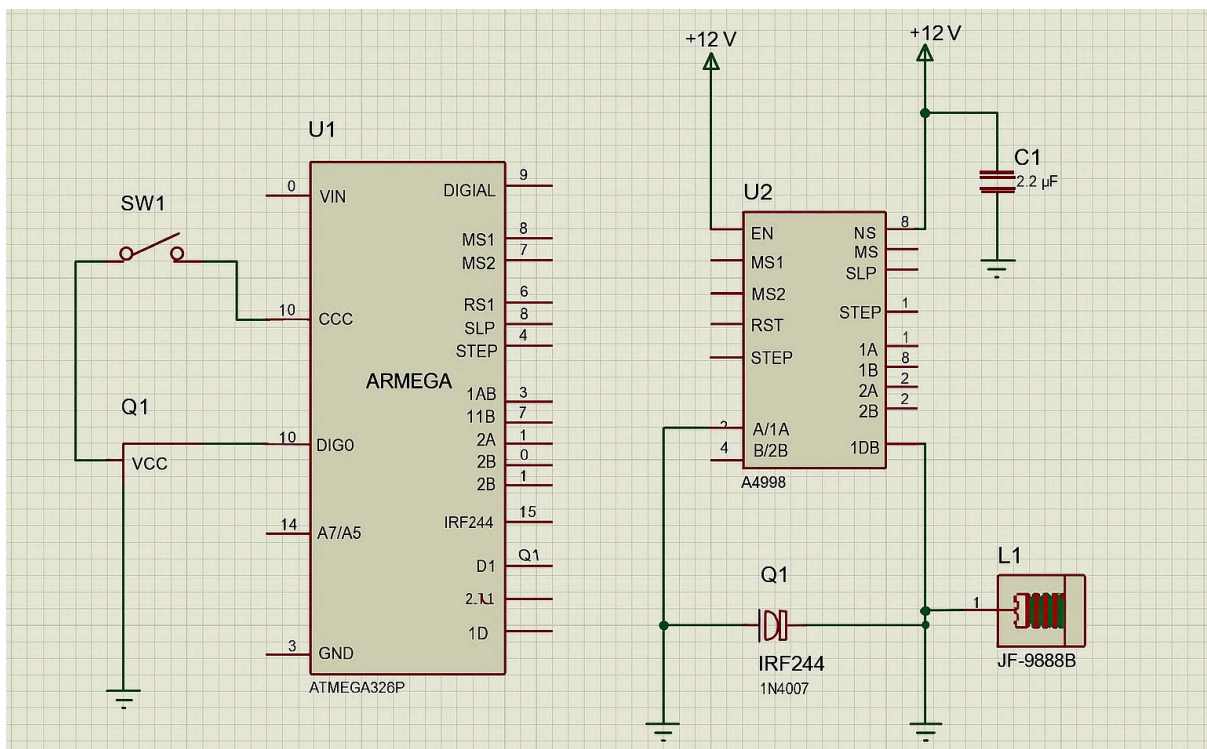


Figure 2.4 Schematic